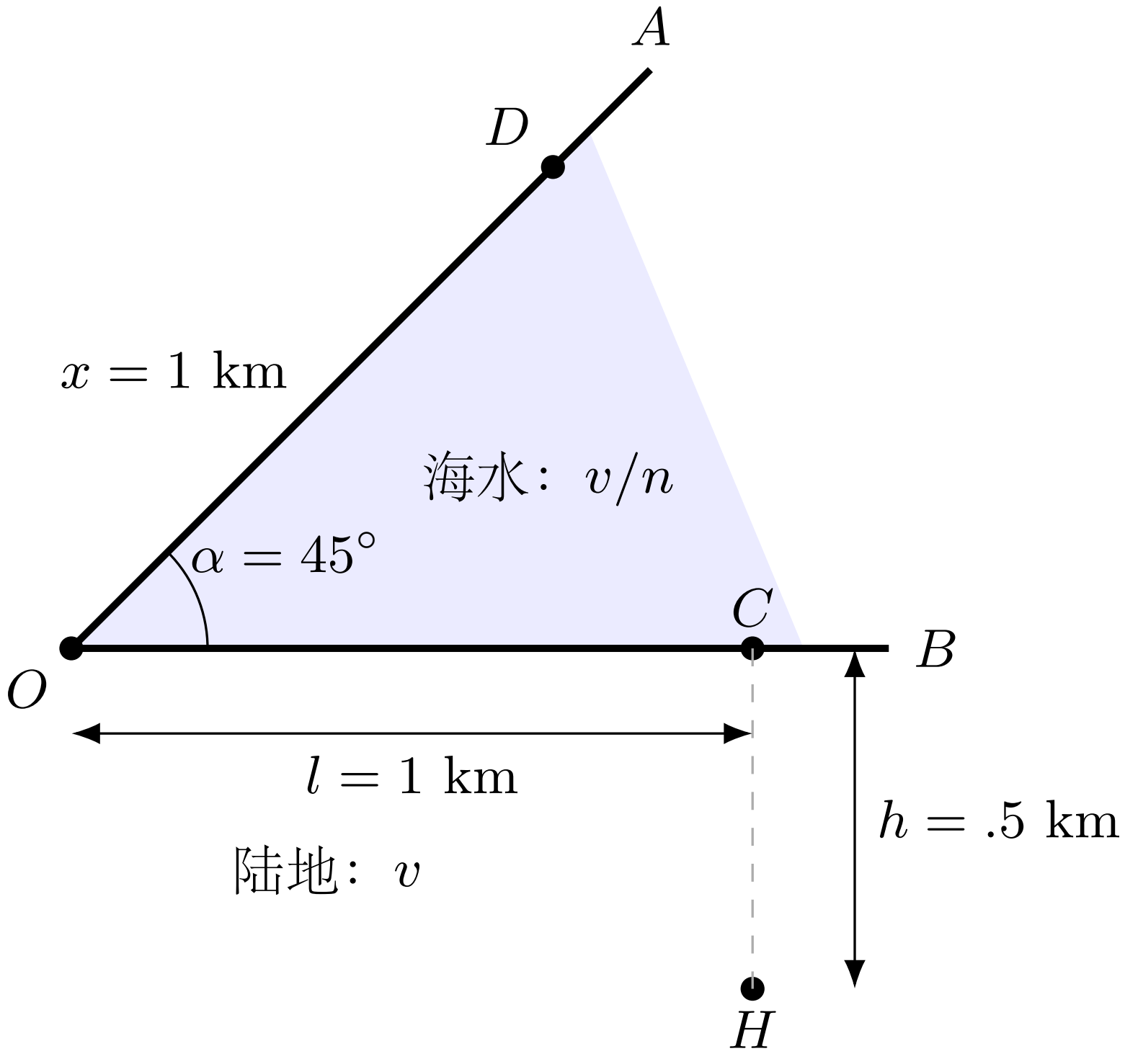
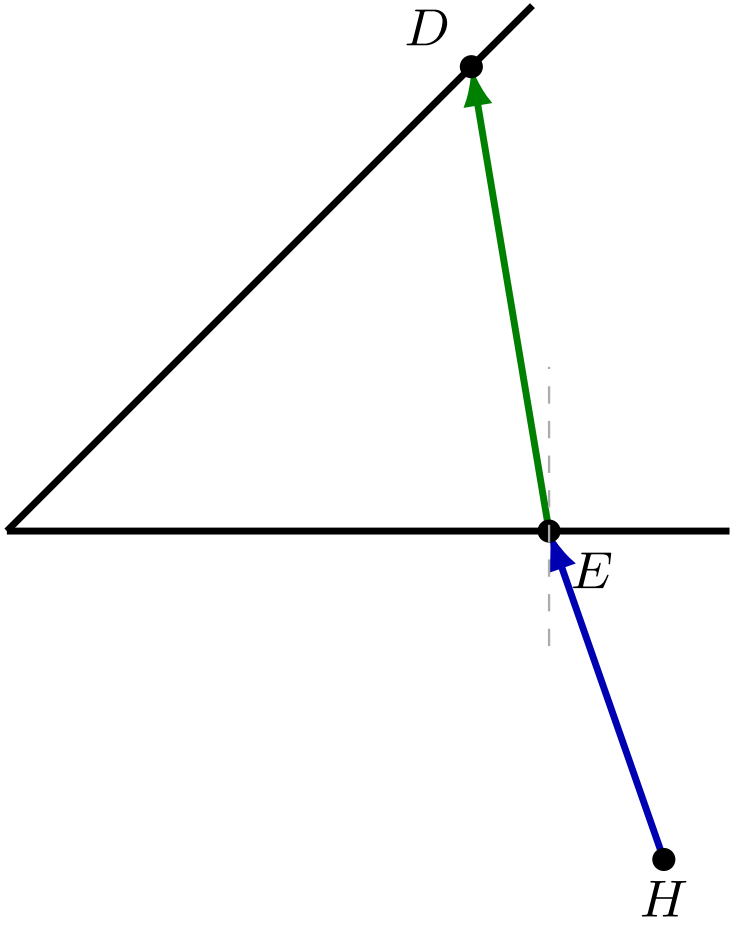
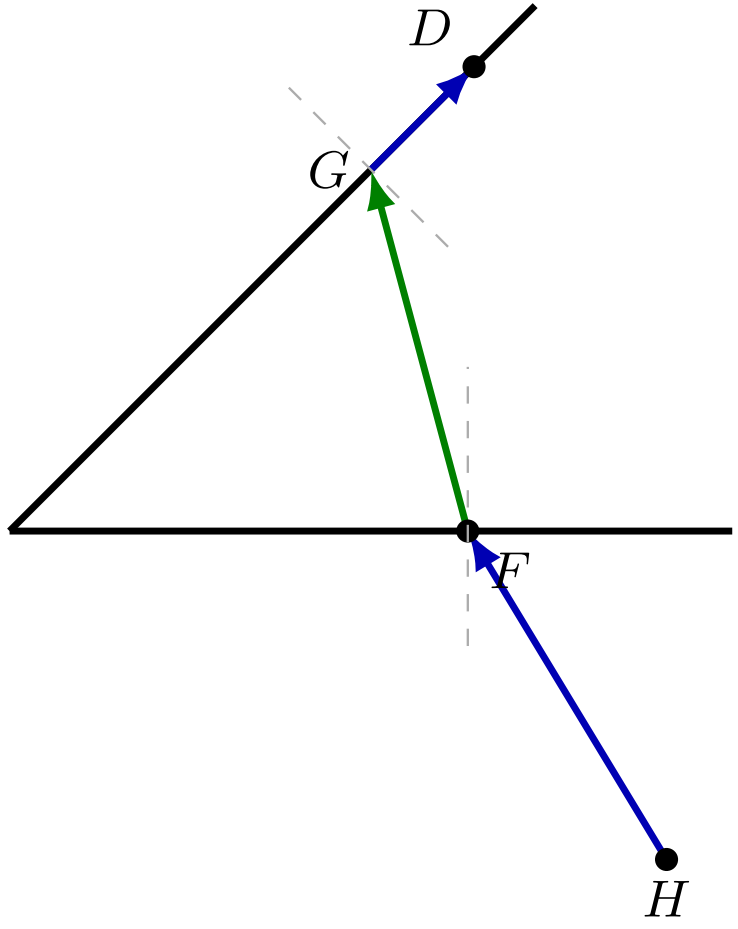

A
D
x = 1 km
海水：v/n
α = 45°
C
B
O
l = 1 km
h = .5 km
陆地：v
H



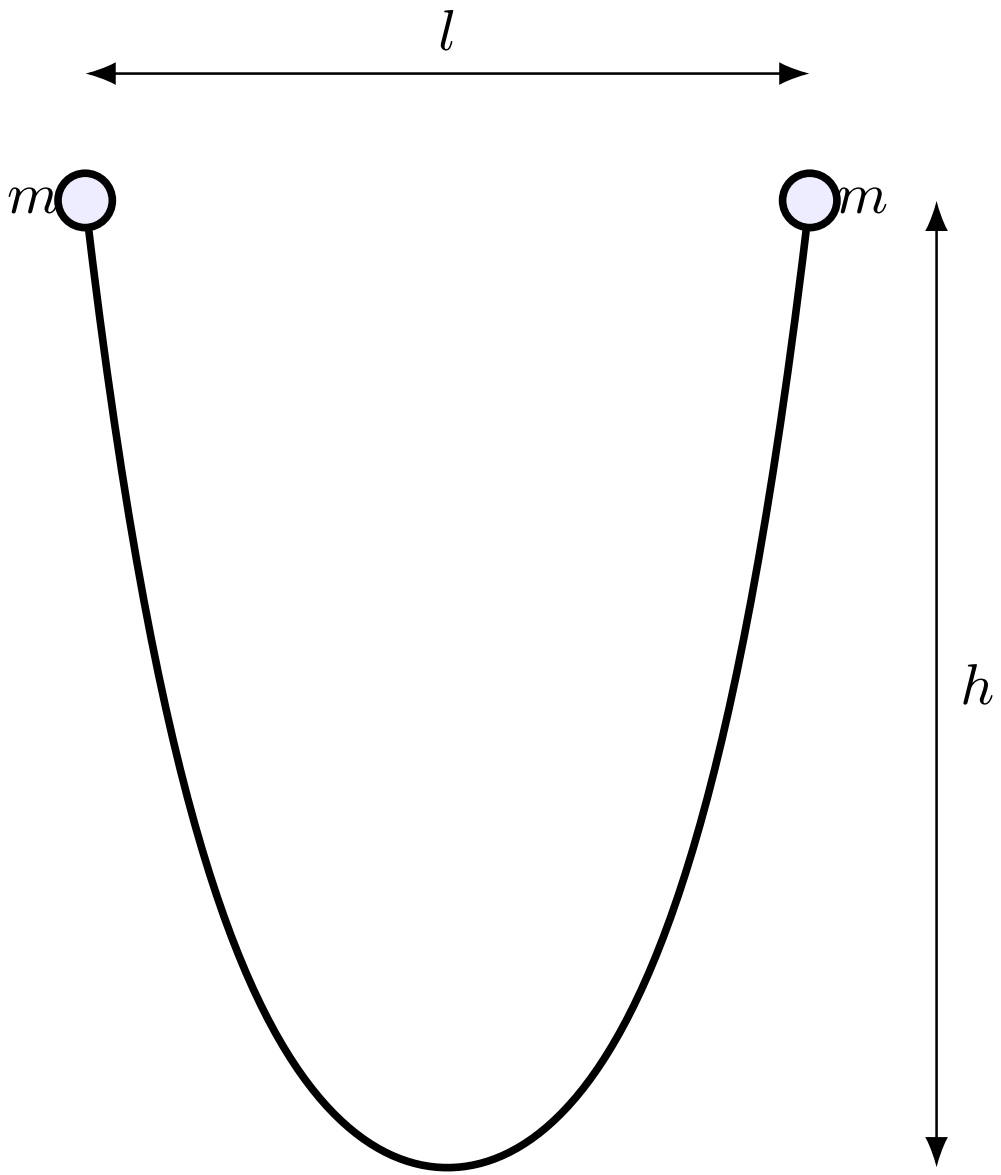
$OE = .82533$ km

$t_2 = 1963.475$ s



$OF = .69750$ km, $OG = .77796$ km

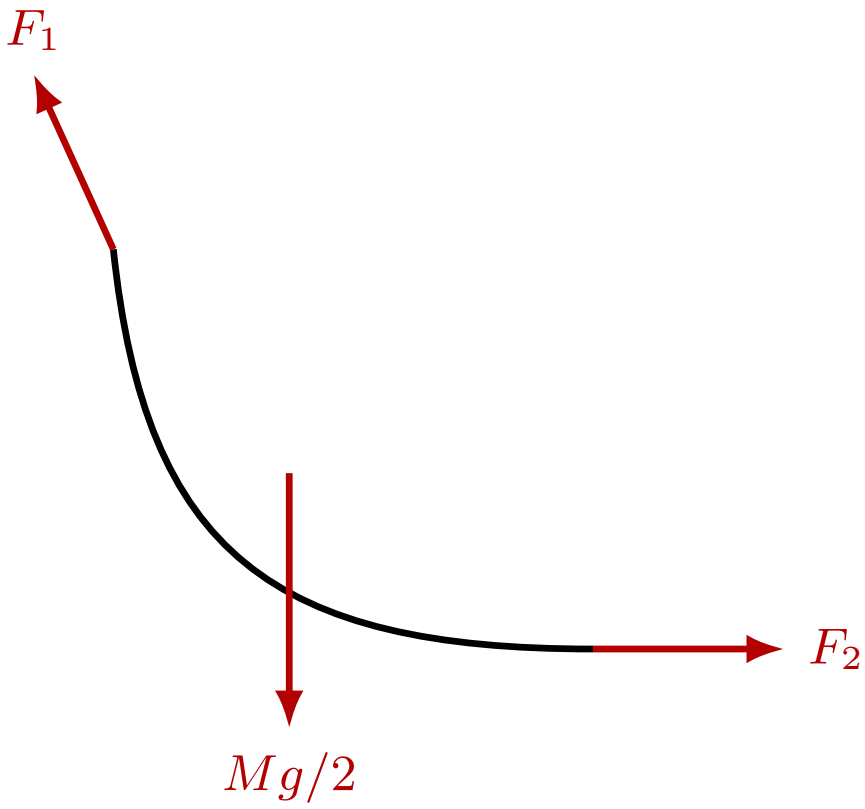
$t_3 = 1945.438$ s



重绳长 $3l$，质量 M

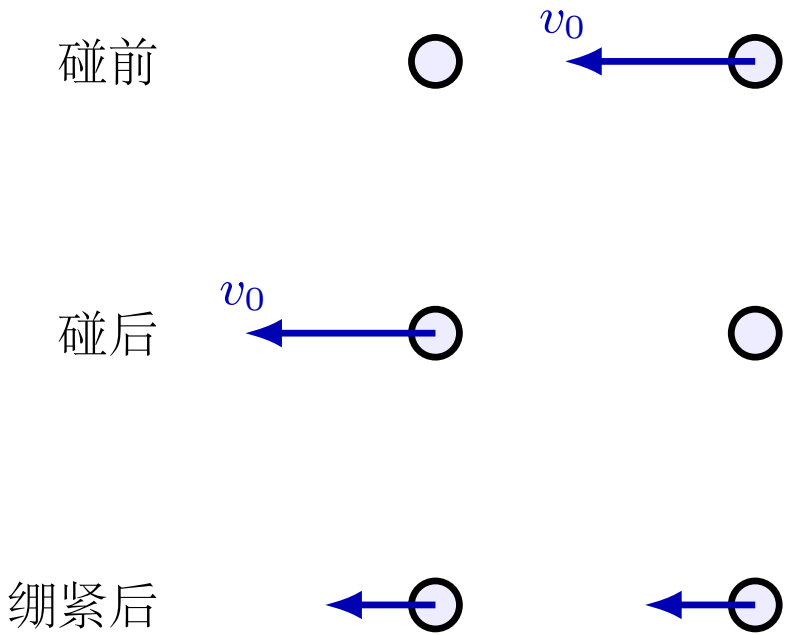
初始位置由支承维持

后续去掉支承，轻绳仍连接两球

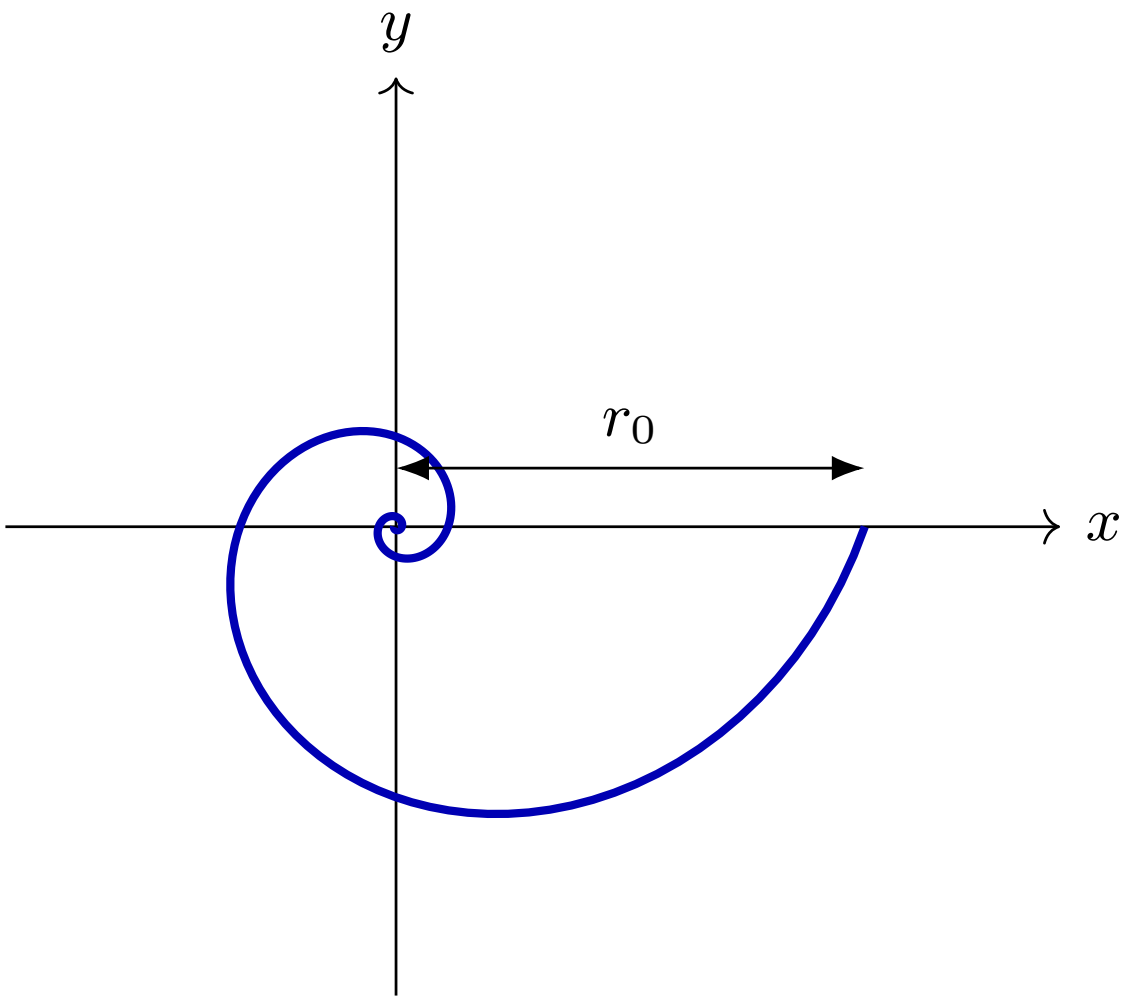


$$F_1^2 - F_2^2 = (Mg/2)^2$$

$$F_1 - F_2 = \lambda g h$$



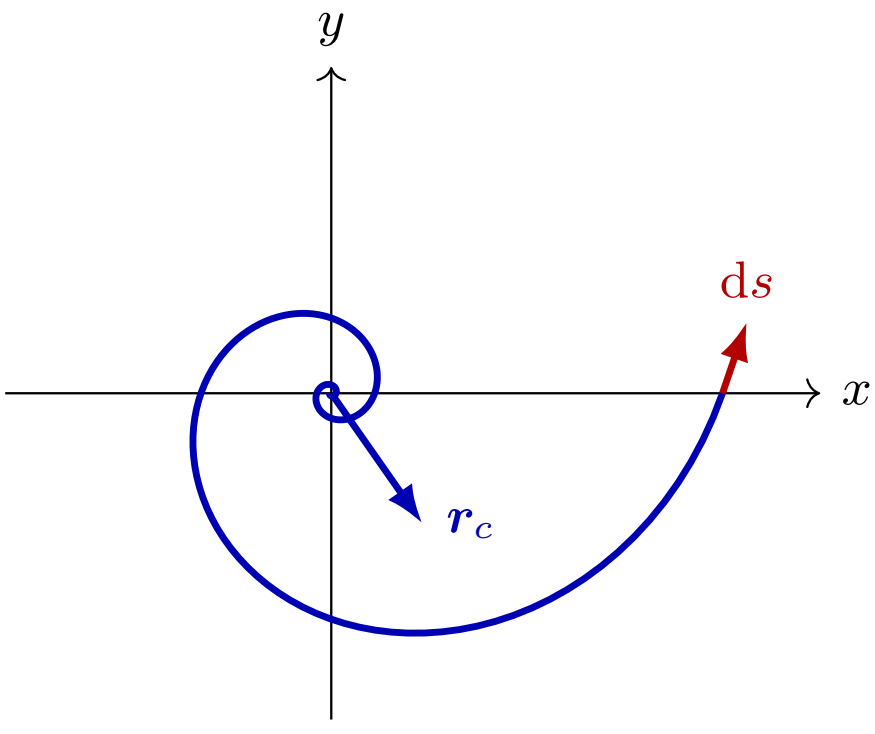
两球均 $-v_0/2$，$J_R = mv_0/2$



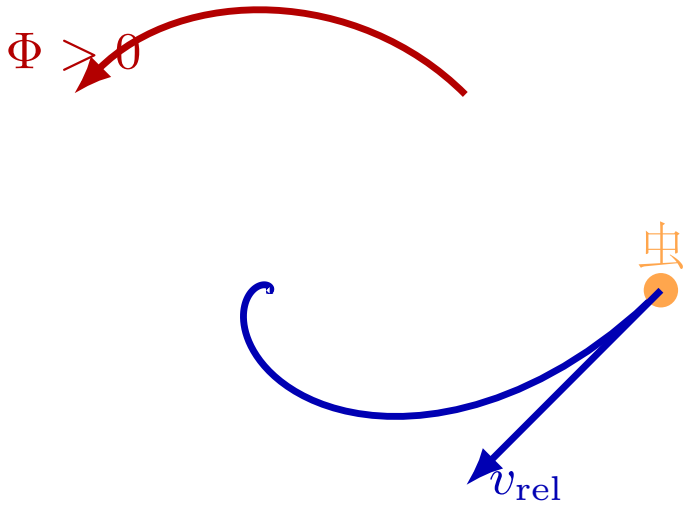
$\rho = r_0 e^{k\theta}$

$\theta \in (-\infty, 0)$，$k > 0$

向内无限绕圈，半径趋零

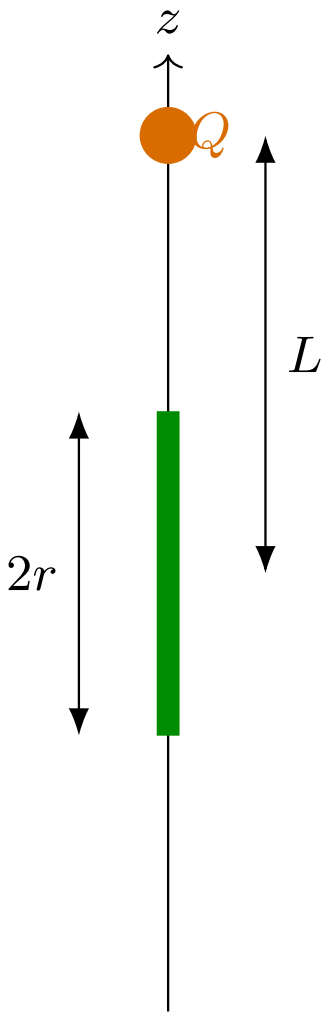


放大 $1 + \mathrm{d}r_0/r_0$ 并旋转 $\mathrm{d}\theta$

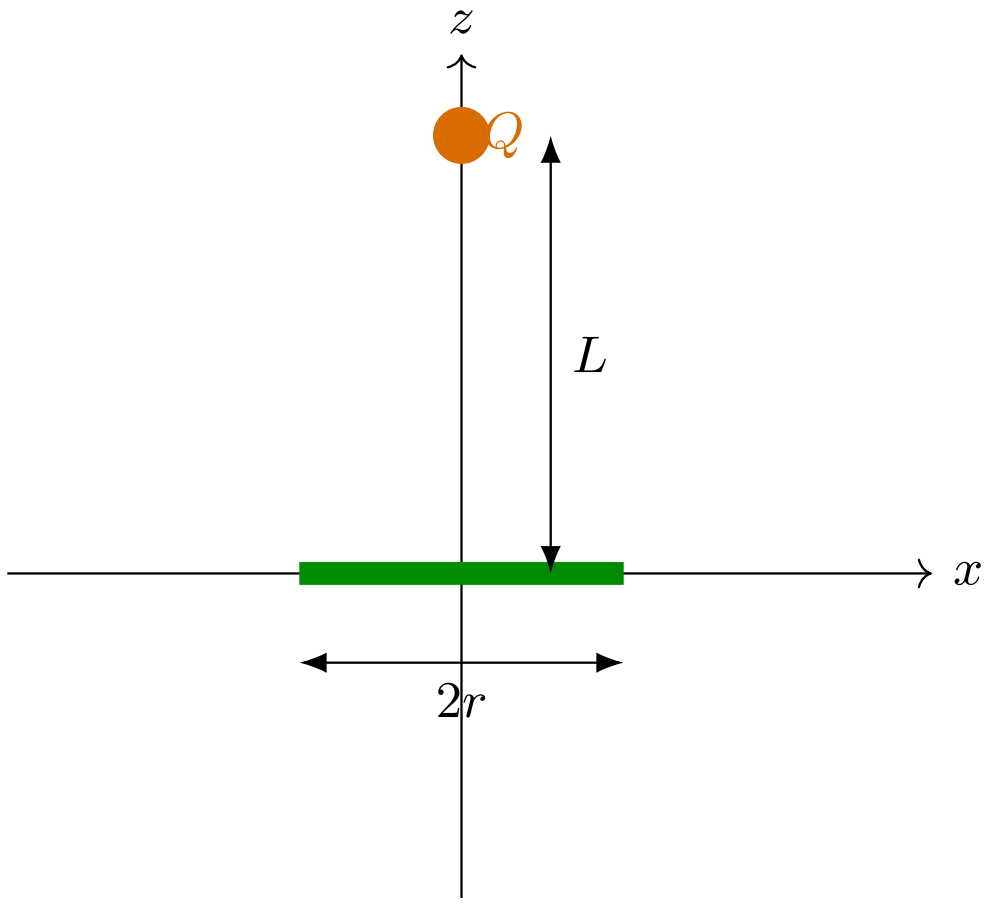


虫向内：$\dot{\theta} < 0$

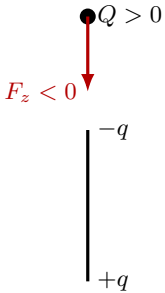
$$I\dot{\Phi} + m_0\rho^2(\dot{\Phi} + \dot{\theta}) = 0$$



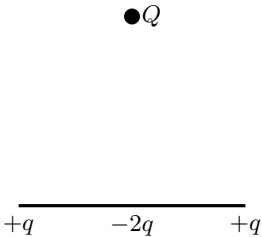
金属棒沿 z 轴



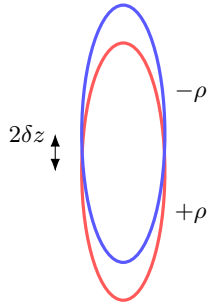
金属棒沿 x 轴



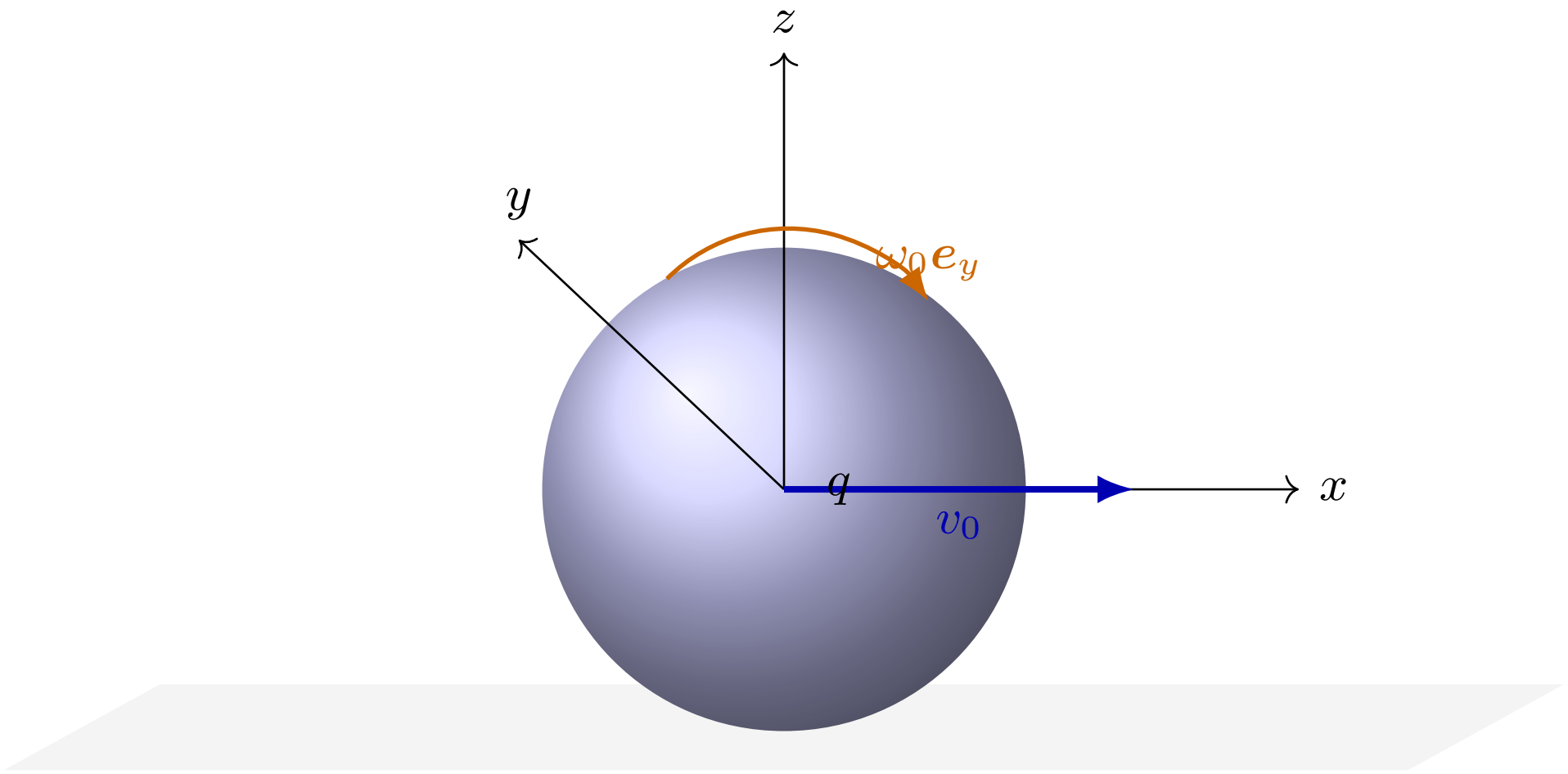
纵向偶极：Q^2r^3/L^5



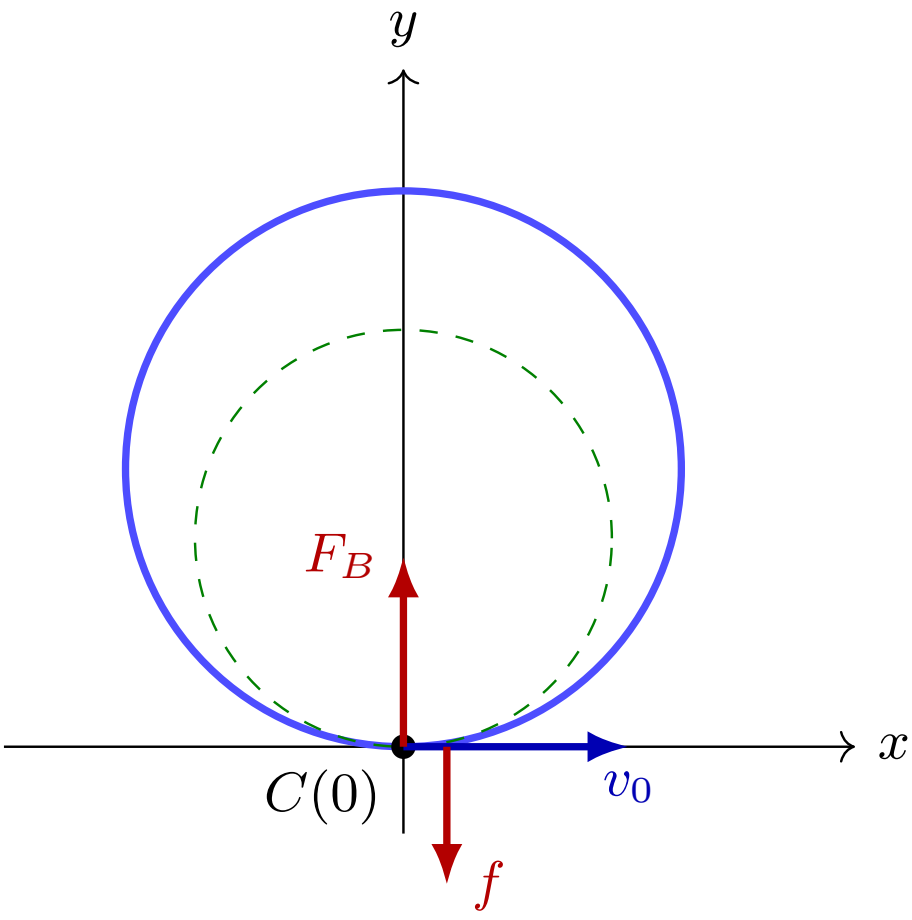
只保留沿棒极化时
横向四极：Q^2r^5/L^7



$\Gamma_2(e) < 0$
保留形状函数，区分力大小与 F_z

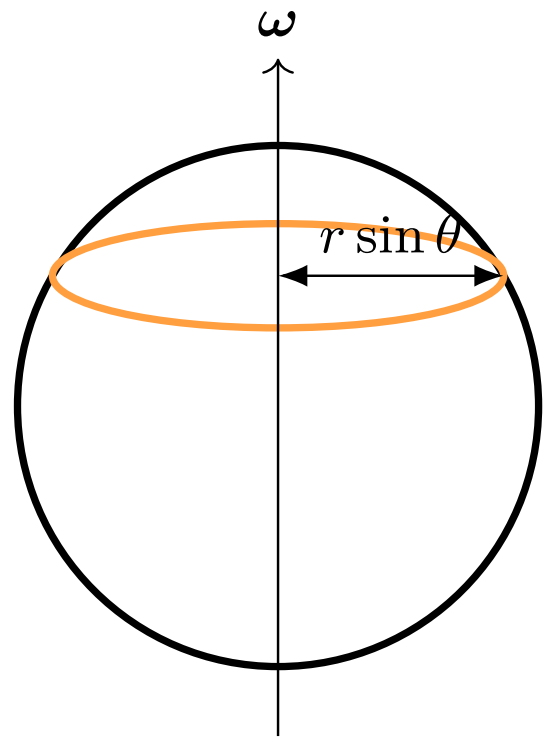


$$\boldsymbol{B} = -B\boldsymbol{e}_z$$



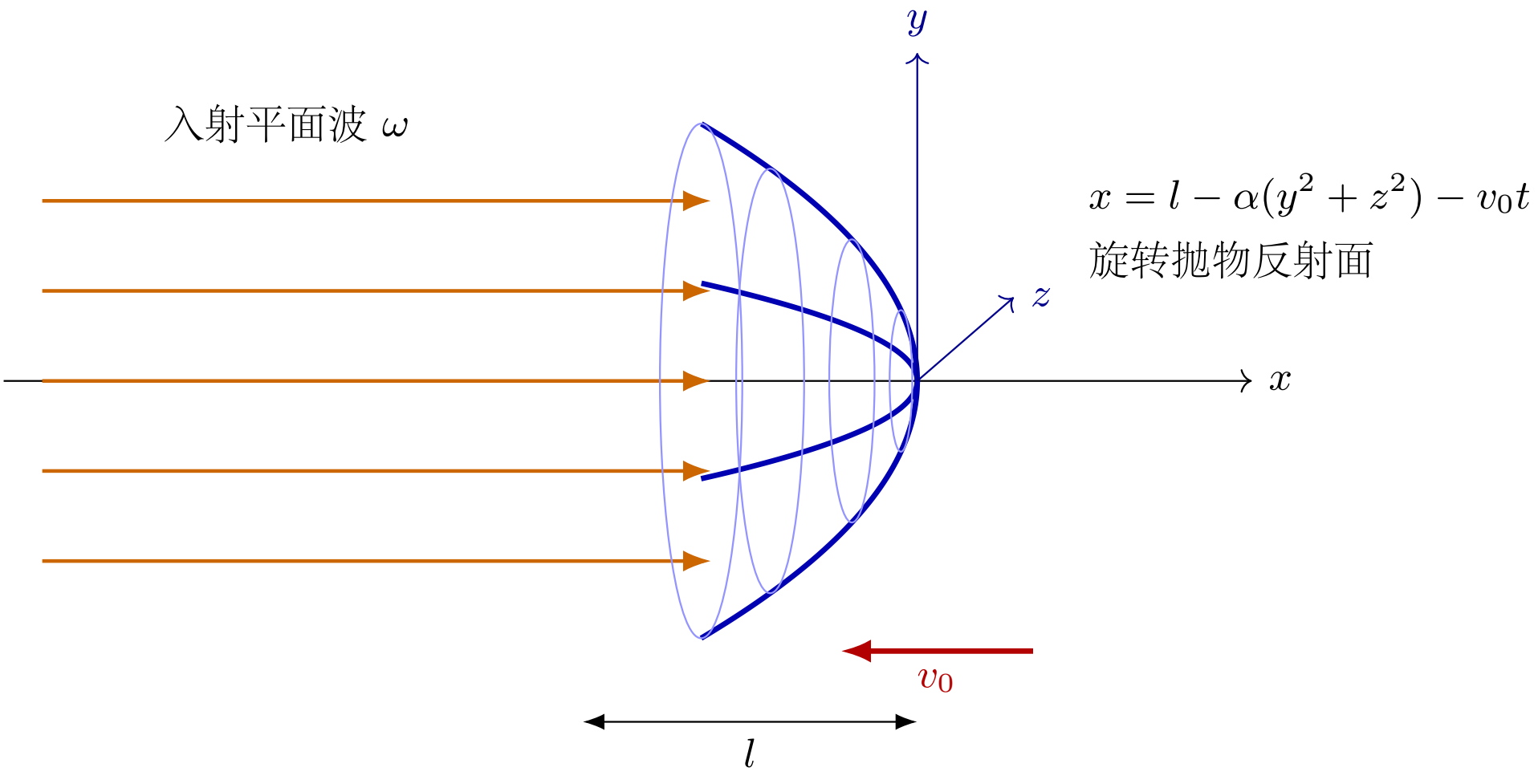
蓝：中心电荷 R_c

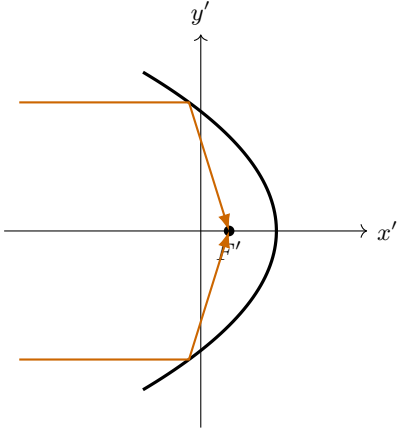
虚线：表面电荷 $R_c' = 3R_c/4$



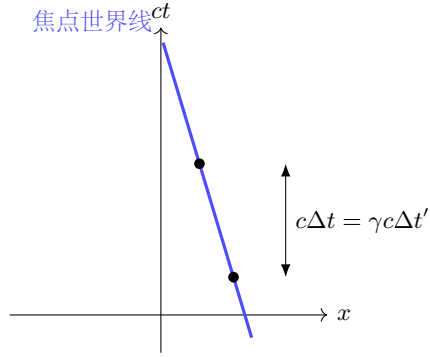
$$\mathrm{d}\mu_m = \mathrm{d}I\,\pi r^2 \sin^2\theta$$

$$\boldsymbol{\mu}_m = (qr^2/3)\boldsymbol{\omega}$$


入射平面波 ω
y
z
x
$x = l - \alpha(y^2 + z^2) - v_0 t$
旋转抛物反射面
v_0
l

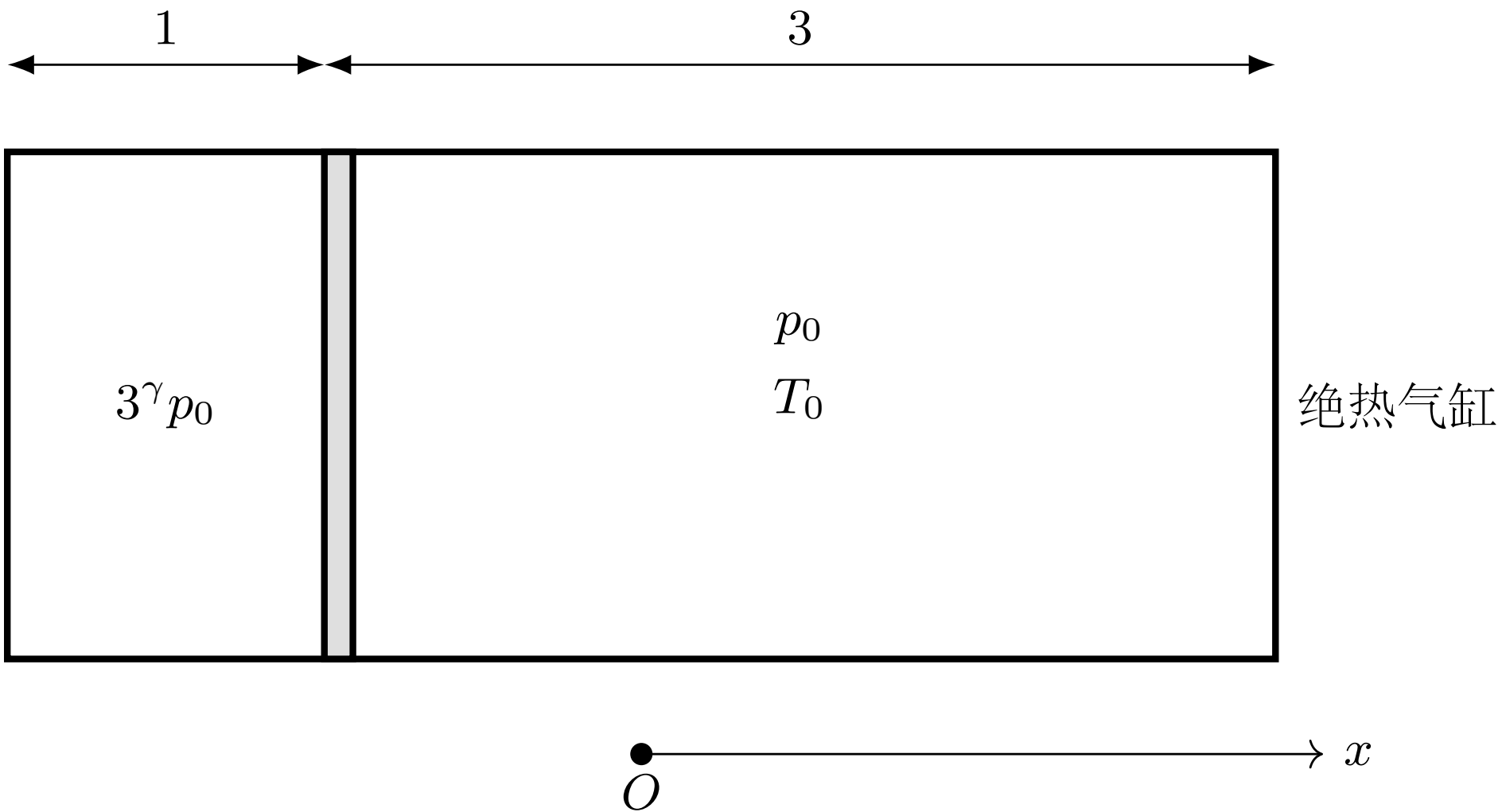


$x'_F = \gamma l - 1/(4\gamma\alpha)$

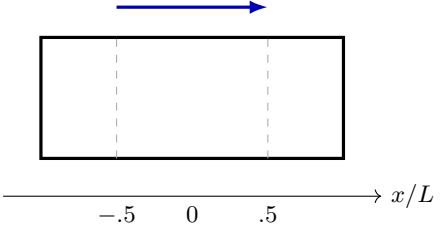


斜率对应 $\mathrm{d}x/\mathrm{d}t = -v_0$

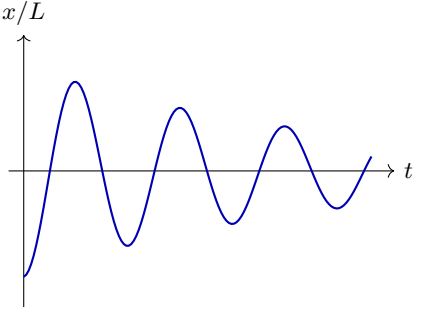
$\omega_1 = \omega'/\gamma = (1 + v_0/c)\omega$


1
3
$3^{\gamma}p_0$
p_0
T_0
绝热气缸
O
x

第一次转折点关于中心对称

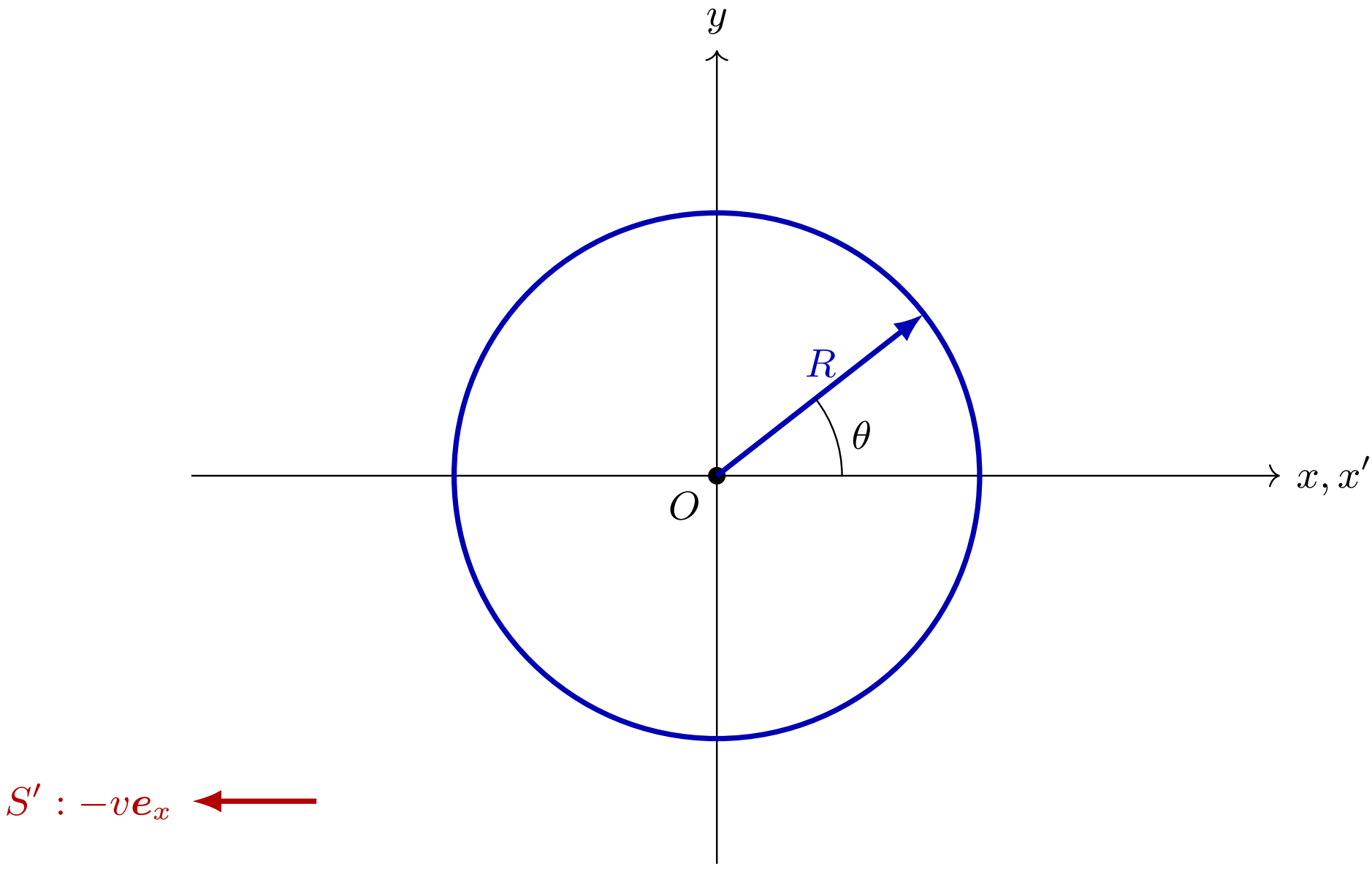


$$p_L V_L^\gamma = p_R V_R^\gamma$$

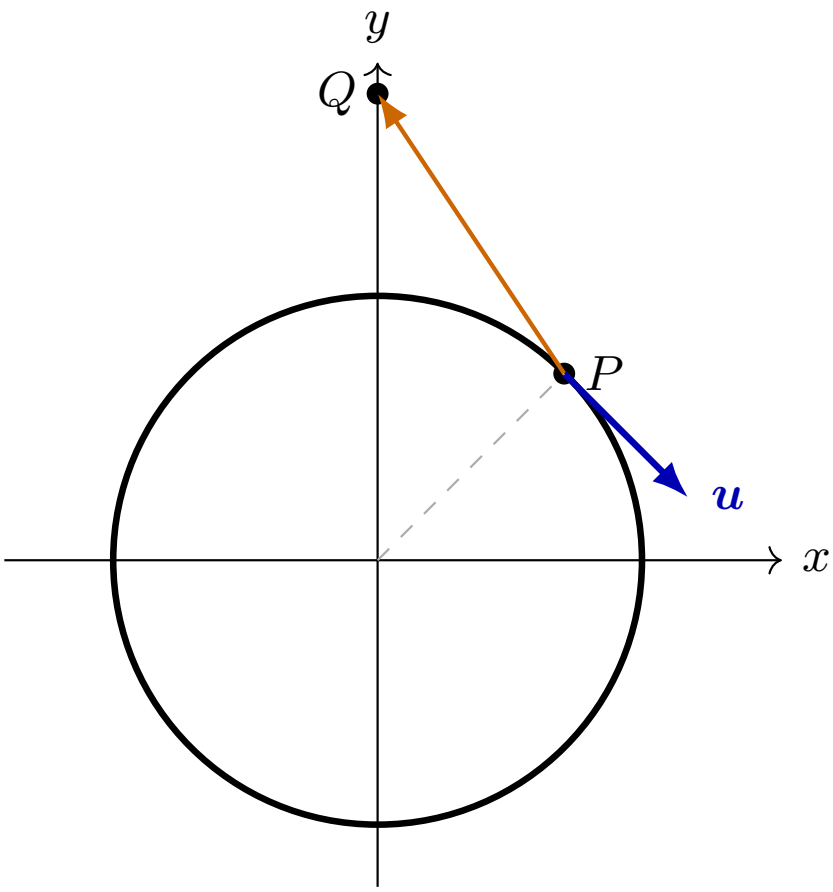


振幅慢变示意；$t_2 \ll t_3$

半振幅时转折点 $x/L \simeq \pm.25$

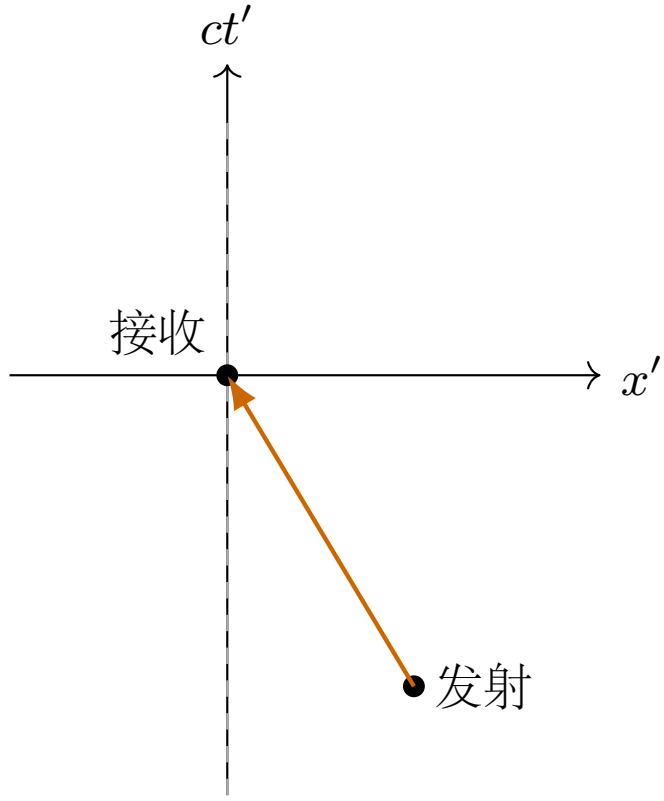


狭义相对论中的转动呼啦圈



顺时针：$\boldsymbol{u} = \omega R(\sin\theta, -\cos\theta)$

光线从 P 指向 Q



接收器 $x' = 0$ 的世界线竖直

发射事件 $t'_P < 0$

图为 x' 投影，光线另有 y' 位移